

FR. CONCEICAO RODRIGUES COLLEGE OF ENGINEERING

Department of Computer Engineering Academic Year 2025-26

Rubrics for Lab Experiments

**Class : B.E. Computer
Engineering**

Subject Name :BDA Lab

Semester : VII

Subject Code :CSL702

Practical No:	8
Title:	Write a program to perform Twitter data/Healthcare data analysis using R language.
Date of Performance:	31-9-25
Roll No:	9913
Name of the Student:	Mark Xavier Lopes

Evaluation:

Performance Indicator	Below average	Average	Good	Excellent	Marks
On time Submission (2)	Not submitted (0)	Submitted after deadline (1)	Early or on time submission(2)	---	
Test cases and output (4)	Incorrect output (1)	The expected output is verified only a for few test cases (2)	The expected output is Verified for all test cases but is not presentable (3)	Expected output is obtained for all test cases. Presentable and easy to follow (4)	
Coding efficiency (2)	The code is not structured at all (0)	The code is structured but not efficient (1)	The code is Structured and efficient. (2)	-	
Knowledge(2)	Basic concepts not clear (0)	Understood the basic concepts (1)	Could explain the concept with suitable example (1.5)	Could relate the theory with real world application(2)	
Total					

Signature of the Teacher:

Experiment No. 8

Aim: Write a program to perform **Twitter data/Healthcare data** analysis using R language.

Objective:

- Understand syntax and environment of Rstudio.
- Use Twitter data/Social data for analysis purpose.
- Perform analysis and plot graphs based on given data

Input and Output:

Any social media dataset will be input

Result: Implementation of the solution for above problem and attach input, program, intermediate and output files.

Conclusion:

Load Required Libraries

```
install.packages(c("tidyverse", "ggplot2", "dplyr", "tm", "wordcloud", "syuzhet", "RColorBrewer"))
```

```
library(tidyverse)
```

```
library(ggplot2)
```

```
library(dplyr)
```

```
library(tm)
```

```
library(wordcloud)
```

```
library(syuzhet)
```

```
library(RColorBrewer)
```

```
# Step 1: Load CSV Dataset
```

```
# -----
```

```
# Upload your CSV to Posit Cloud and replace "healthcare_tweets.csv" with your file name
```

```
tweets <- read.csv("healthcare_tweets.csv", stringsAsFactors = FALSE)
```

```
# Preview first few rows
```

```
head(tweets)
```

```
# -----
```

```
# Step 2: Clean Text Data
```

```
# -----
```

```
tweets$text <- gsub("http\\S+|www\\S+|@\\S+|#\\S+", "", tweets$text) # Remove URLs, mentions,
hashtags
```

```
tweets$text <- gsub("\\s+", " ", tweets$text) # Remove extra spaces
```

```
# -----
```

```
# Step 3: Word Frequency Analysis
```

```
# -----
```

```
corpus <- Corpus(VectorSource(tweets$text))
```

```
corpus <- tm_map(corpus, content_transformer(tolower))
```

```
corpus <- tm_map(corpus, removePunctuation)
```

```
corpus <- tm_map(corpus, removeNumbers)
```

```
corpus <- tm_map(corpus, removeWords, stopwords("english"))
```

```
dtm <- TermDocumentMatrix(corpus)
```

```
m <- as.matrix(dtm)
```

```
word_freq <- sort(rowSums(m), decreasing = TRUE)
```

```
df <- data.frame(word = names(word_freq), freq = word_freq)
```

```

# Top 10 words

head(df, 10)

# -----

# Step 4: Sentiment Analysis

# -----

tweets$sentiment <- get_nrc_sentiment(tweets$text)

sentiment_summary <- colSums(tweets$sentiment[,1:8])

sentiment_summary

# -----

# Step 5: Visualizations

# -----

# Word Cloud

wordcloud(words = df$word, freq = df$freq, min.freq = 2,
          max.words = 100, random.order = FALSE,
          colors = brewer.pal(8, "Dark2"))

# Sentiment Bar Plot

barplot(sentiment_summary, las = 2, col = rainbow(8),
        main = "Sentiment Analysis of Tweets")

# -----

# Step 6: Save Outputs

# -----

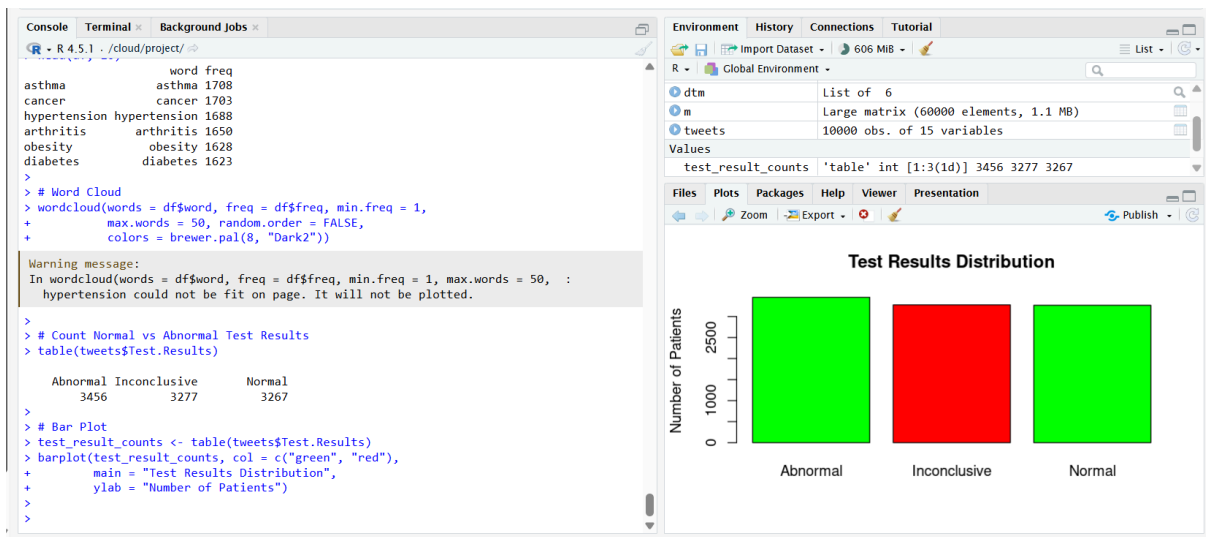
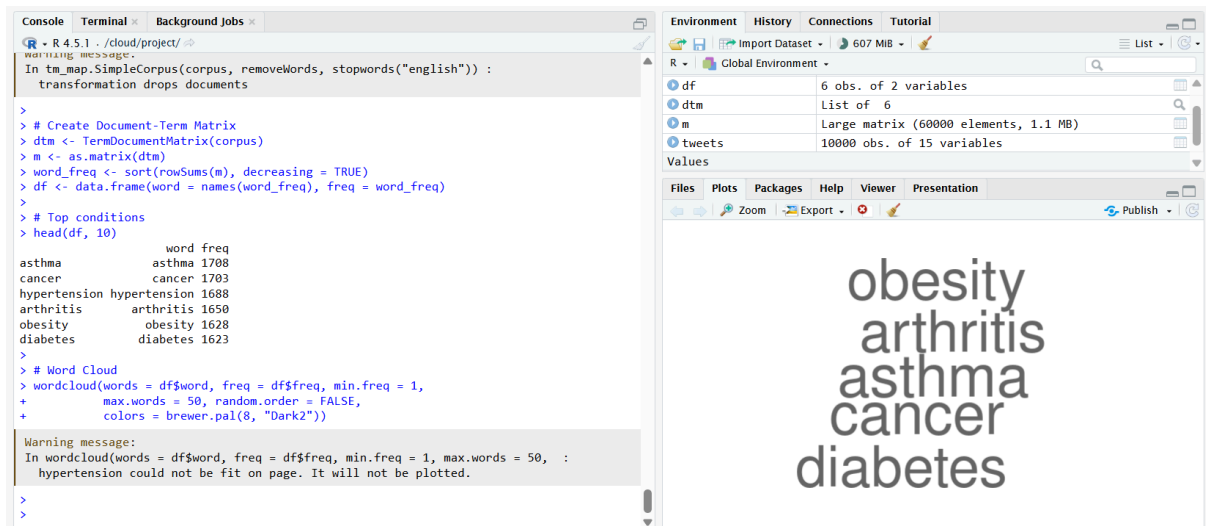
write.csv(tweets, "cleaned_tweets.csv", row.names = FALSE)

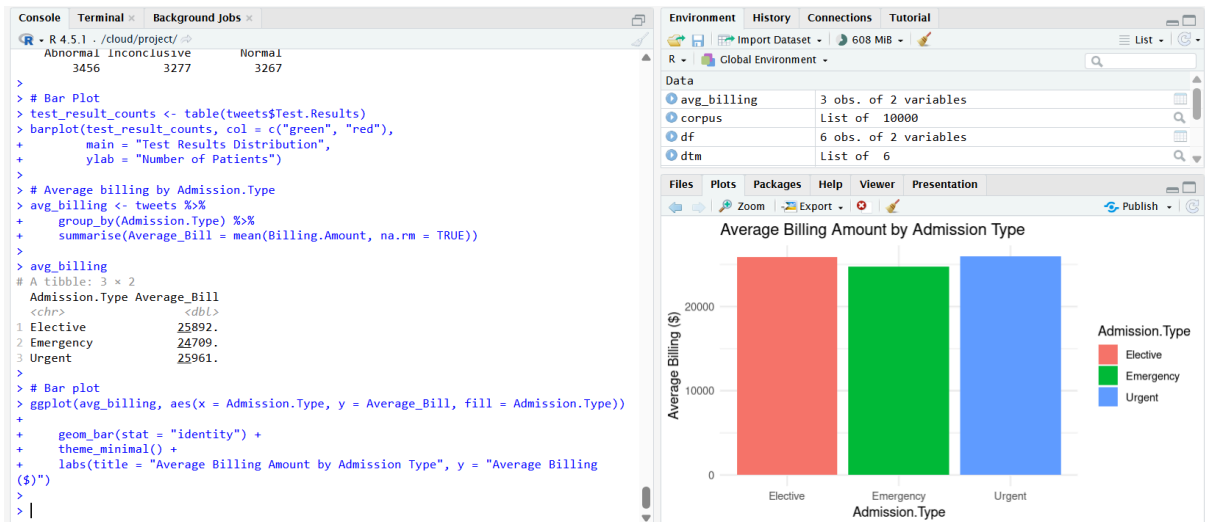
```

```
write.csv(df, "word_frequency.csv", row.names = FALSE)
```

```
write.csv(as.data.frame(sentiment_summary), "sentiment_summary.csv")
```

Analysis

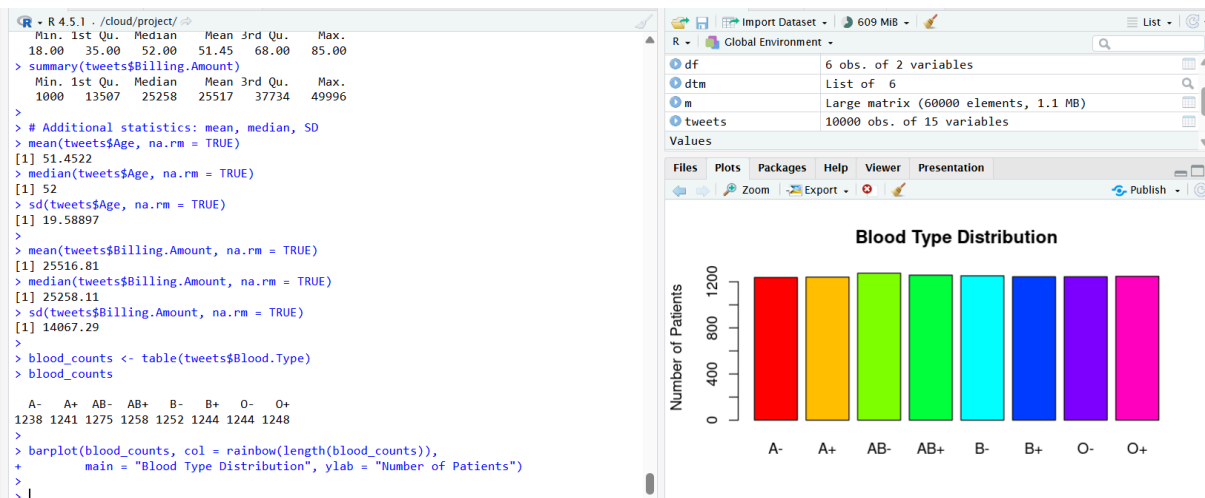




```

> ggplot(avg_billing, aes(x = Admission.type, y = Average_Bill, fill = Admission.type))
+
+   geom_bar(stat = "identity") +
+   theme_minimal() +
+   labs(title = "Average Billing Amount by Admission Type", y = "Average Billing
($)")
>
> # Summary for Age and Billing.Amount
> summary(tweets$Age)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  18.00  35.00   52.00   51.45  68.00   85.00
> summary(tweets$Billing.Amount)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  1000  13507   25258   25517  37734   49996
>
> # Additional statistics: mean, median, SD
> mean(tweets$Age, na.rm = TRUE)
[1] 51.4522
> median(tweets$Age, na.rm = TRUE)
[1] 52
> sd(tweets$Age, na.rm = TRUE)
[1] 19.58897
>
> mean(tweets$Billing.Amount, na.rm = TRUE)
[1] 25516.81
> median(tweets$Billing.Amount, na.rm = TRUE)
[1] 25258.11
> sd(tweets$Billing.Amount, na.rm = TRUE)
[1] 14067.29
>
> |

```



POSTLAB: